

Create keys and certificates

- Release version: Australia
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- ⌚ 2 minutes to read

Create keys and certificates in your root directory to enable Transport Layer Security (TLS) setup. TLS setup is necessary before you can configure mTLS on the MID Web Server and agent.

Before you begin

- Ensure that there are no keys installed in the MID unified keystore by running the following command:

```
bin/scripts/manage-certificates.(sh/bat) -l
```

When there are no keys installed, the output is: `defaultsecuritypairhandle`

- Invalidate your MID Server.

If additional keys exist in the output, reinstall those keys after invalidating the MID Server.

- Ensure that the MID Server is connected to the instance.
- Select a directory in which you want to create certificates, to be called the root directory.

Note:

The commands specified in the following procedure are relevant only for a Centos7 host. When working with another OS, use the commands relevant for your host.

Role required: agent_client_collector_admin

Procedure

1. In your root directory, create subdirectories for your certificates.

```
mkdir -p labca labmid labacc;
```

2. In your root directory:
 - a. Generate a custom certificate authority key pair.

```
openssl ecparam -list_curves;
openssl ecparam -out labca/ec-labcakey.pem -name prime256v1 -genkey;
ls labca/;
```

The generated output is the ec-labcakey.pem file.

b. Run the following commands:

```
openssl ecparam -in labca/ec-labcakey.pem -text -noout;
openssl req -x509 -new -nodes -key labca/ec-labcakey.pem -sha512 -day
openssl verify labca/labcacert.pem;
```

The generated output is:

- labca/labcacert.pem: C = <country>, ST = <state>, L = <location>, O = <organization>, OU = <organization unit>, CN = <cn abbreviation>
- error 18 at 0 depth lookup: self signed certificate
- OK

Note:

The error message can be ignored.

3. Prepare the MID Web Server key and certificate.

a. Run the following commands in the root directory:

```
sudo cp -a labca/labcacert.pem /etc/pki/ca-trust/source/anchors/;
sudo update-ca-trust extract;
openssl verify labca/labcacert.pem
```

The generated output is: labca/labcacert.pem: OK

b. Run the command "hostname --all-fqdns" to obtain all of the valid hostnames for the specific VM.

c. Run the following commands:

```
openssl req -new -newkey rsa:4096 -keyout labmid/rsa-labmidkey.pem -
openssl x509 -req -days 365 -in labmid/mid.csr -CA labca/labcacert.pe
```

Enter the FQDN as the <hostname> value in the previous command. If more than one fqdn value is returned by the command, use the value with the following format: hostname.domain.domain.com .

The generated output is: Signature ok subject=/C=<country>/ST=<state>/L=<location>/O=<organization>/OU=<organization unit>/CN=<mid server host fqdn>: Getting CA Private Key

4. In your root directory, combine the key and cert files into one file, named mid.pem.

```
cat labmid/rsa-labmidkey.pem labmid/mid.crt > labmid/mid.pem;
```

What to do next

Install the .pem file in the MID unified keystore and set up the MID Web Server.